1. A specific organisms can always be found in association with a given disease. If we take typhoid as an example it is caused by a bacterium *Salmonella typhi.*
2. The organism can be isolated and grown in pure culture in the laboratory. *Salmonella typhi* are grown in soild media under laboratory conditions.
3. The pure culture will produce the diseases when inoculated into a susceptible animal.

Almost all the pathogenic organisms produce the same disease in experimental animals. Usually rats, mice, rabbits or guinea pigs are used as experimental animals. *Pneumococci* produce pneumonia in animals. *Salmonella* species do not produce typhoid fever in rat, mice or rabbit. So chimpanzee is taken as experimental animal and it produces fever in chimpanzee.

4. It is possible to recover the organism in pure culture from the experimentally infected animals and it is observed to be the same as originally inoculated pathogen. Figure 1.7 explains the Koch's postulates.

Limitations

Some organisms have not yet been grown in artificial culture media

Example: *Mycobacterium leprae* and *Treponema pallidum*.

Modern addition to Koch's Postulates

Today we recognize additional criteria of causal relation between a microorganism and a disease. The important one is the demonstration of abnormally high concentration of specific circulating antibodies to the organism in the infected host or the presence of abnormally high degree of specific immunity or hypersensitivity to the infecting agent in

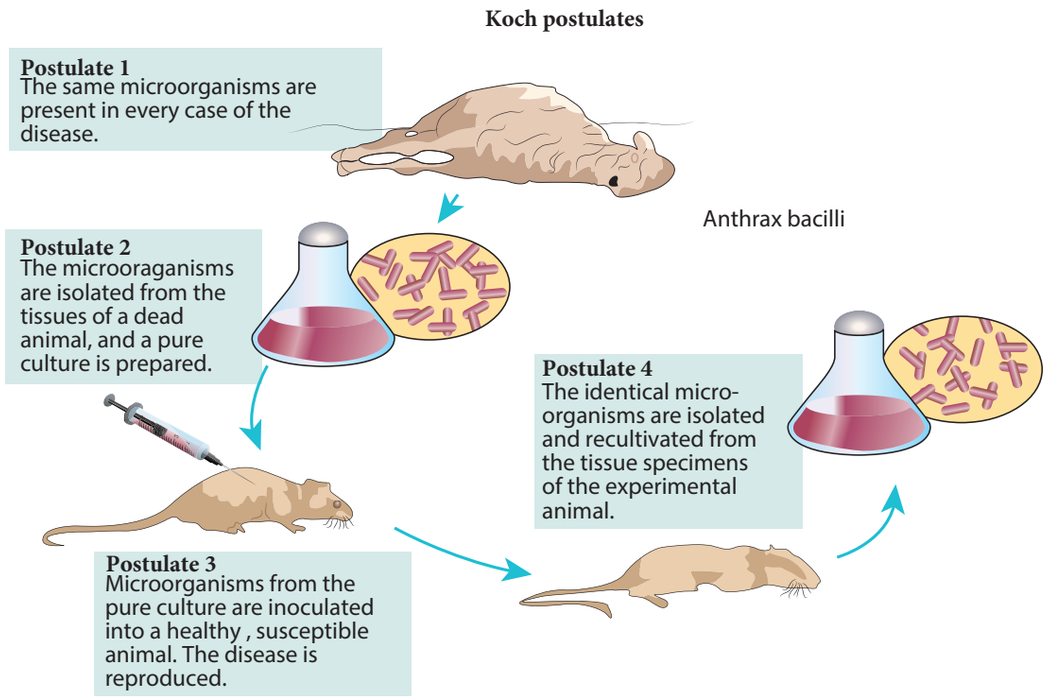


Figure 1.7: Koch's postulates for infectious diseases

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